

Instituto Tecnológico de Costa Rica

Operations Research - Semester II

Simplex

Members:

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	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	-2	-3	0	0	-1M	-1M	0
	0	0.5	0.25	1	0	0	0	4
	0	1	3	0	-1	1	0	20
	0	1	1	0	0	0	1	10

Intermediate Tables

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	2M	4M	0	-1M	0	0	30M
	0	0.5	0.25	1	0	0	0	4
	0	1	3	0	-1	1	0	20
	0	1	1	0	0	0	1	10

Pivoting 1

Most Positive

Column 3 (3999997.00)

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	2M	4M	0	-1M	0	0	30M
	0	0.5	0.25	1	0	0	0	4
	0	1	3	0	-1	1	0	20
	0	1	1	0	0	0	1	10

Fractions

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b	Frac
	1	2M	4M	0	-1M	0	0	30M	
	0	0.5	0.25	1	0	0	0	4	16.00
	0	1	3	0	-1	1	0	20	6.67
	0	1	1	0	0	0	1	10	10.00

$$4.00/0.25 = 16.00$$

$$20.00/3.00 = 6.67$$

$$10.00/1.00 = 10.00$$

Smallest fraction: 6.67 $\rightarrow$ Pivot: row 3

Pivot

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	2M	4M	0	-1M	0	0	30M
	0	0.5	0.25	1	0	0	0	4
	0	1	3	0	-1	1	0	20
	0	1	1	0	0	0	1	10

Canonization

$$R_3 \leftarrow R_3/3.00$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	2M	4M	0	-1M	0	0	30M
	0	0.5	0.25	1	0	0	0	4
	0	0.333	1	0	-0.333	0.333	0	6.67
	0	1	1	0	0	0	1	10

$$R_1 \leftarrow R_1 + -3999997.00R_3$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.5	0.25	1	0	0	0	4
	0	0.333	1	0	-0.333	0.333	0	6.67
	0	1	1	0	0	0	1	10

$$R_2 \leftarrow R_2 + -0.25R_3$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.417	0	1	0.0833	-0.0833	0	2.33
	0	0.333	1	0	-0.333	0.333	0	6.67
	0	1	1	0	0	0	1	10

$$R_4 \leftarrow R_4 + -1.00R_3$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.417	0	1	0.0833	-0.0833	0	2.33
	0	0.333	1	0	-0.333	0.333	0	6.67
	0	0.667	0	0	0.333	-0.333	1	3.33

Pivot Result

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.417	0	1	0.0833	-0.0833	0	2.33
	0	0.333	1	0	-0.333	0.333	0	6.67
	0	0.667	0	0	0.333	-0.333	1	3.33

Pivoting 2

Most Positive

Column 2 (666665.67)

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.417	0	1	0.0833	-0.0833	0	2.33
	0	0.333	1	0	-0.333	0.333	0	6.67
	0	0.667	0	0	0.333	-0.333	1	3.33

Fractions

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b	Frac
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M	
	0	0.417	0	1	0.0833	-0.0833	0	2.33	5.60
	0	0.333	1	0	-0.333	0.333	0	6.67	20.00
	0	0.667	0	0	0.333	-0.333	1	3.33	5.00

$$2.33/0.42 = 5.60$$

$$6.67/0.33 = 20.00$$

$$3.33/0.67 = 5.00$$

Smallest fraction: 5.00 $\rightarrow$ Pivot: row 4

Pivot

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.417	0	1	0.0833	-0.0833	0	2.33
	0	0.333	1	0	-0.333	0.333	0	6.67
	0	0.667	0	0	0.333	-0.333	1	3.33

Canonization

$$R_4 \leftarrow R_4/0.67$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0.667M	0	0	0.333M	-1.33M	0	3.33M
	0	0.417	0	1	0.0833	-0.0833	0	2.33
	0	0.333	1	0	-0.333	0.333	0	6.67
	0	1	0	0	0.5	-0.5	1.5	5

$$R_1 \leftarrow R_1 + -666665.67R_4$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0	0	0	-0.5	-1M	-1M	25
	0	0.417	0	1	0.0833	-0.0833	0	2.33
	0	0.333	1	0	-0.333	0.333	0	6.67
	0	1	0	0	0.5	-0.5	1.5	5

$$R_2 \leftarrow R_2 + -0.42R_4$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0	0	0	-0.5	-1M	-1M	25
	0	0	0	1	-0.125	0.125	-0.625	0.25
	0	0.333	1	0	-0.333	0.333	0	6.67
	0	1	0	0	0.5	-0.5	1.5	5

$$R_3 \leftarrow R_3 + -0.33R_4$$

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0	0	0	-0.5	-1M	-1M	25
	0	0	0	1	-0.125	0.125	-0.625	0.25
	0	0	1	0	-0.5	0.5	-0.5	5
	0	1	0	0	0.5	-0.5	1.5	5

Pivot Result

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0	0	0	-0.5	-1M	-1M	25
	0	0	0	1	-0.125	0.125	-0.625	0.25
	0	0	1	0	-0.5	0.5	-0.5	5
	0	1	0	0	0.5	-0.5	1.5	5

Results

	Z	x_1	x_2	s_1	e_1	a_1	a_2	b
	1	0	0	0	-0.5	-1M	-1M	25
	0	0	0	1	-0.125	0.125	-0.625	0.25
	0	0	1	0	-0.5	0.5	-0.5	5
	0	1	0	0	0.5	-0.5	1.5	5

Objective Value

$$Z = 25.00$$

Variables

$$x_1 = 5.00$$

$$x_2 = 5.00$$

Slack or Surplus

$$s_1 = 0.25$$

$$s_2 = 0.00$$

$$s_3 = 0.00$$